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Hyperbaric Oxygen Therapy Reverses Periodontitis-Associated Porphyromonas gingivalis–Induced Cognitive Impairment in Sprague–Dawley Rats

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ABSTRACT

Porphyromonas gingivalis (P. gingivalis), a Gram-negative obligate anaerobe, as a major pathogen in periodontitis, it has been implicated in the development of neurodegenerative disorders through systemic inflammatory mechanisms. Chronic exposure to P. gingivalis and its virulence factors—lipopolysaccharide (LPS) and gingipain—can compromise the integrity of the blood-brain barrier (BBB), increase amyloid-β (Aβ) accumulation, and trigger neuroinflammation, contributing to cognitive deficits such as impaired spatial working memory (SWM). In this experimental study, male Sprague-Dawley rats were bilaterally injected with "P. gingivalis (1×10^9 CFU/mL)" into the mandibular gingiva for five weeks to induce periodontitis and cognitive dysfunction. Hyperbaric Oxygen Therapy (HBOT), an environmentally based therapeutic approach that modifies tissue oxygen conditions, was then administered at 2.0 ATA for 60 minutes per day, five days a week, over four weeks. The spontaneous alternation test using a Y-maze was employed to evaluate SWM at three timepoints: baseline (pre-induction), after bacterial induction, and after HBOT treatment. The results showed that HBOT significantly improved cognitive performance in Sprague Dawley with P. gingivalis-induced impairment. These findings support the potential of HBOT as a promising adjunctive therapy for addressing the neurological consequences of chronic periodontal infection. This further underscores the role of HBOT in mitigating neurodegenerative risks associated with periodontitis.

KEYWORDS: Soleus muscle exercise, Weight, Type 2 diabetes mellitus, Blood glucose

INTRODUCTION

A key contributor to the development of periodontitis, a long-term inflammatory disorder of the tooth-supporting tissues, is the Gram-negative anaerobic bacterium *Porphyromonas gingivalis* (P. gingivalis) [1]. P. gingivalis produces virulence factors (gingipains, LPS) that damage periodontal tissue and trigger immune overreaction, leading to alveolar bone resorption and tooth loss [2]. *P. gingivalis* infection is not confined to the oral cavity but is also associated with various systemic diseases, including cognitive disorders, through mechanisms such as inflammatory spillover and microbiome dysbiosis [3].

Cognitive impairment is marked by reductions in attention, memory, and executive abilities, and its severity spans from Mild Cognitive Impairment (MCI) to serious disorders including dementia and Alzheimer's disease [4].

In contrast, gingipains, as proteolytic enzymes characteristic of P. gingivalis, can compromise tissue integrity by degrading host proteins, including structural components of epithelial and endothelial barriers, thereby increasing blood–brain barrier permeability [5].

The combined activity of LPS and gingipain not only triggers a localized inflammatory response in gingival tissues but also facilitates the translocation of P. gingivalis along with its virulence factors into the systemic circulation, leading to bacteremia. Within the bloodstream, LPS binds to soluble CD14 molecules and activates endothelial cells and monocytes, while gingipain modulates the complement system and coagulation pathways. This interaction results in a low-grade systemic inflammatory state that contributes to the disruption of blood-brain barrier function [6,7].

Due to systemic inflammation and epithelial–endothelial barrier disruption induced by LPS and gingipain, *P. gingivalis* can access the central nervous system via three main pathways: (1) disruption of tight intercellular junctions form a critical component of the blood-brain barrier caused by gingipain's proteolytic activity [8]; (2) transcellular transport of LPS mediated by TLR4 activation; and (3) retrograde migration through cranial nerves, primarily the trigeminal nerve [16]. These mechanisms facilitate bacterial invasion into brain tissue and trigger neuroinflammatory processes, thereby reinforcing the pathogenic link between periodontal infection and cognitive disorders.

MATERIAL AND METHODS

2.1 Animal and experimental protocol

In this study, the experimental subjects were divided into two groups: a control group and a treatment group, each consisting of 11 male Sprague–Dawley rats, 10 weeks old, weighing between 150 and 180 g. Before receiving treatment, all rats were acclimated for one week under regulated conditions, including a temperature range of 18–26 °C, relative humidity of 30–70%, and a 12-hour light–dark cycle. They were kept in standard laboratory cages with free access to pellet diet and drinking water.

Inclusion criteria encompassed healthy status confirmed by veterinary examination, as well as normal physiological characteristics such as smooth fur, pink eyes, normal activity levels, good appetite, firm feces, and normal social behavior (group interactions). Exclusion criteria included congenital abnormalities identified by a consulting veterinarian. Animals that died during the experimental process were classified as dropouts and excluded from the final analysis.

2.2 Injection of Pg-LPS into the gingiva

After the acclimatization period was complete, the experimental animals in the treatment group underwent induction of Porphyromonas gingivalis (ATCC 33277 PK/5) to trigger experimental periodontitis. Induction was performed with an infection dose of 3 µl (1×10^9 CFU/ml) using a 5 µl microsyringe (Hamilton, Switzerland) applied to the second molar teeth on the right and left sides of the lower jaw. This procedure was repeated every 2 days for 5 consecutive weeks, based on a modified method from Jin et al. (2023) [29].

2.3 Hyperbaric Oxygen Therapy (HBOT)

After induction, Sprague-Dawley rats underwent hyperbaric oxygen therapy (HBOT) in a chamber pressurized to 2.0 ATA with 100% medical oxygen. The HBOT protocol consisted of three phases: (1) gradual compression at a rate of 0.1 ATA/min; (2) core therapy lasting 60 minutes, divided into two 30-minute sessions with a 5-minute interval at the target pressure; and (3) slow decompression at 0.1 ATA/min to prevent barotrauma. Prior to therapy, animals were fasted for 4–6 hours and underwent physical condition assessment. During HBOT, animal activity was monitored through the chamber window, with session termination upon signs of severe distress. After therapy, animals were transferred to a recovery cage for 30 minutes. This protocol was conducted five times per week for four weeks [28]. All procedures received ethical approval from the institutional animal research ethics committee. This HBOT protocol also represents an environmental-based approach, since it modifies oxygen levels in the tissue microenvironment.

2.4 Y-Maze Test

The Spontaneous Alternation Test (SAT) was conducted using a Y-maze apparatus consisting of three identical arms (A, B, and C) arranged at 120° angles, each measuring approximately 30 cm × 8 cm × 15 cm, to evaluate spatial working memory in rats. Prior to testing, Sprague-Dawley rats were acclimated to the testing room for 10–15 minutes to minimize stress. At the start of the test, each rat was gently placed at the end of one arm and allowed to freely explore all three arms for 5–8 minutes. All activity was recorded via video camera to document the sequence and number of entries into each arm. An entry was considered valid when the entire body of the rat, including the tail, fully entered the arm.

At the end of each session, researchers recorded both the total arm entries and the instances of spontaneous alternations, defined as consecutive entries into three distinct arms. The spontaneous alternation percentage was then determined using the formula: (spontaneousalternations/totalentries–2)×100. To eliminate residual odor cues, the Y-maze was wiped with 70% alcohol between sessions before introducing the next subject. The assessment was carried out at three different time points: prior to *P. gingivalis* induction (pretest), at week 5 (post-induction), and at week 9 (post-HBOT). [9].

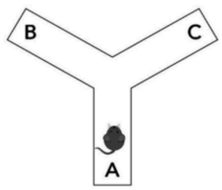

Figure 1. The Y-maze used in this study, with a dimension 30 cm × 8 cm × 15 cm and 120° between the two legs, made of polyvinyl chloride (PVC) and assembled using PVC glue [48].

RESULT

This study investigates the effects of hyperbaric oxygen therapy (HBOT) following *Porphyromonas gingivalis* (PG) induction, assessed through changes in spatial working memory measured by the Y-maze spontaneous alternation test at weeks 1, 5, and 9. The collected data were tabulated using descriptive statistics and subsequently analyzed with inferential statistical tests at a 95% significance level ($p = 0.05$). Data analysis was performed using IBM SPSS

Table 1. Differences Test for Control Group (Paired t-Test Results)

	K_Pretest	K_Week 5	K_Week 9
K_Pretest		0.063	0.069
K_Week 5			0.231
K_Week 9			

Table 2. Mean and Standard Deviation of Y-Maze Results

Groups		n	Mean ± Standard Deviation
Control	Pretest	11	70.85 ± 4.97
	Week 5	11	65.45 ± 9.6
	Week 9	11	68 ± 5.08
Treatment	Pretest	11	68.02 ± 1.67
	Week 5	11	32.01 ± 24.71
	Week 9	11	52.09 ± 14.03

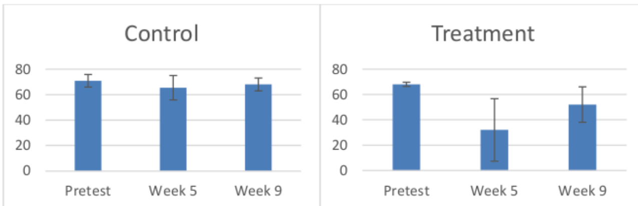

Figure 2. Mean and Standard Deviation of Y-Maze Results

Statistics version 27.0.
The results indicated that in the control group, no significant changes were observed in spatial working memory over the study period. In contrast, the treatment group exhibited a decline in spatial working memory following P. gingivalis induction at week 5, which subsequently improved after administration of hyperbaric oxygen therapy (HBOT).

The results obtained the Shapiro-Wilk test was applied to examine normality, while the Levene test was used to assess homogeneity. The statistics showed that the control group was normally distributed and homogeneous (P >0.05), but the treatment group data was not normally distributed (P<0.05). Therefore, the examination of variance performed on the control group was the Repeated Measures ANOVA test to test all three and the Paired T-Test to test each group. Meanwhile, for the difference test of the treatment group, a non-parametric statistical test was used, namely the Friedman test to test all three and the Wilcoxon test to test each group.
Since all sig. values > 0.05, there is no significant difference between the pretest in Week 5 and Week 9 in the control group.
Since all sig. values are < 0.05, there is a significant difference between the Week 5 and Week 9 pre-tests in the treatment group.

DISCUSSION

This study used male Sprague Dawley rats that were first induced with Porphyromonas gingivalis for five weeks to establish a chronic inflammatory condition capable of impairing cognitive function. Following this induction period, the rats received hyperbaric oxygen therapy (HBOT) at 2 ATA for four consecutive weeks. Changes in spatial working memory were assessed at each treatment stage using the Y-maze Spontaneous Alternation Test to evaluate the potential of HBOT in restoring spatial memory deficits induced by P. gingivalis exposure. HBOT exerts its therapeutic effects through oxygen modulation, contributing to the reduction of neuroinflammation and oxidative stress.

In the treatment group, there was a decrease in spatial working memory in the pretest group compared to week 5 after intraoral induction of Pg. This finding is consistent with the study by Mamunur et al. (2023), which showed that intraperitoneal exposure to Pg-LPS caused a significant decrease in the percentage of Spontaneous Alternation Behavior (SAB) in the Y-maze test compared to the control group. This decline indicates impaired spatial working memory function due to inflammatory stimulation by P. gingivalis. This impairment is suspected to occur because P. gingivalis or its virulence products, such as lipopolysaccharide (LPS) and gingipain, can reach brain tissue by

Table 3. Differences Test for Treatment Group (Wilcoxon Signed-Rank Test Results)

	K_Pretest	**K_Week 5**	**K_Week 9**
K_Pretest		<0.005	<0.010
K_Week 5			<0.041
K_Week 9			

crossing the blood–brain barrier (BBB) and triggering a neuroinflammatory response [15,30].

Chronic infection by Porphyromonas gingivalis, a gram-negative anaerobic bacterium and the primary pathogen in periodontitis, begins with colonization in the oral cavity, particularly within periodontal tissues. This bacterium produces various virulence factors, including lipopolysaccharide (LPS), gingipain, fimbriae, and outer membrane vesicles (OMVs), which facilitate invasion into gingival tissues and penetration into the bloodstream (bacteremia). The condition is exacerbated by epithelial tissue damage and the formation of periodontal pockets, which increase bacterial access to local blood vessels [31].

Upon entering the systemic circulation, "*P. gingivalis* and its virulence components can stimulate activation of the innate immune system. The interaction of P. gingivalis lipopolysaccharide (LPS) with receptors such as Toll-like receptor 2 (TLR2) and C5aR1 on immune cells triggers activation of the PI3K and NF-κB pathways, which subsequently induce the release of systemic proinflammatory cytokines, including IL-1β, TNF-α, IL-6, and IL-8" [32,33]. These inflammatory mediators not only sustain local inflammation but also disseminate through the bloodstream to distant organs such as the brain, heart, and liver, thereby eliciting a systemic inflammatory response.

Systemic dissemination of proinflammatory cytokines, particularly TNF-α and IL-1β, plays a critical role in disrupting the blood–brain barrier (BBB). One of the underlying mechanisms involves the downregulation of tight junction proteins, such as occludin and claudin-5, in brain capillary endothelial cells. These alterations increase BBB permeability to large molecules and immune cells, thereby facilitating the entry of toxins, cytokines, and pathogens into brain tissue. This condition triggers microglial activation and exacerbates neuroinflammation [34,35].

In addition to blood circulation, transmission of P. gingivalis to the central nervous system can also occur via sensory nerve pathways, particularly the trigeminal nerve. Outer membrane vesicles (OMVs) released by this bacterium are able to migrate retrograde through the sensory branches of the trigeminal nerve to the trigeminal nucleus and hippocampus. This mechanism forms part of the oral–brain axis and contributes to reinforcing the link between chronic periodontal infection and neuroinflammatory pathogenesis [36,37].

The migration route of Porphyromonas gingivalis to the central nervous system can occur directly or indirectly. Directly, P. gingivalis infection increases blood–brain barrier (BBB) permeability through modulation of membrane transport protein expression and endothelial structural protein expression. One of the main mechanisms is the reduction in expression of Major Facilitator Superfamily Domain Containing 2a (Mfsd2a), a transporter protein that maintains BBB integrity by inhibiting vesicular transcytosis pathways in brain microvascular endothelial cells. The reduction in Mfsd2a due to P. gingivalis exposure triggers increased expression of Caveolin-1 (Cav-1), a key protein in caveolae formation on the plasma membrane. Activation of Cav-1 accelerates caveolae formation and enhances transcytosis activity, allowing foreign molecules to more easily penetrate the BBB and increase permeability. Indirectly, P. gingivalis releases outer membrane vesicles (OMVs) containing lipopolysaccharides (LPS), gingipain, and fimbriae. These OMVs are able to evade immune system detection, cross the BBB, and trigger a neuroinflammatory response even though intact bacteria do not migrate to brain tissue [38].

Upon reaching the brain, lipopolysaccharide (LPS) from *P. gingivalis* induces increased production of reactive oxygen species (ROS) through the activation of NADPH oxidase 4 (NOX4) in microglial cells, accompanied by elevated levels of interleukin-6 (IL-6) and interleukin-8 (IL-8). This process reflects a degenerative neuroinflammatory mechanism that contributes to reduced neuronal viability and hyperphosphorylation of tau protein [39]. In addition, the release of proinflammatory cytokines such as IL-1β, IL-6, and tumor necrosis factor-α (TNF-α), together with ROS from peripheral tissues during chronic infection, can enter the systemic circulation and cross the blood–brain barrier (BBB) when its permeability is increased due to endotoxin exposure or oxidative stress. The presence of these inflammatory mediators in the central nervous system activates microglia, amplifies the neuroinflammatory response, and leads to decreased neuronal viability. Collectively, these processes ultimately contribute to the decline in cognitive function, particularly spatial working memory [40,41].

At the same time, gingipain, a virulent protease from P. gingivalis, not only directly damages periodontal tissue but also disrupts immune pathways and hemostasis, potentially crossing the blood-brain barrier and causing neurological effects. Gingipain is known to cleave tau protein, leading to hyperphosphorylation and the formation of neurofibrillary tangles (NFTs), one of the hallmark features of Alzheimer's disease [4]. This finding reinforces the link between chronic periodontitis and neurodegenerative disorders through both systemic inflammatory and molecular pathways.

A study by Dominy et al. (2019) also showed that lipopolysaccharide (LPS) and gingipain, two major virulence factors of P. gingivalis, were successfully identified in brain specimens from Alzheimer's Disease (AD) patients. These two factors trigger the activation of microglia/macrophages through the NF-κB and Cathepsin B pathways, which in turn increase the production of Aβ1-42/40. The accumulation of Aβ deposits in the brain and exacerbates the pathology of Alzheimer's disease [4].

Experimental data from the present study revealed an improvement in spatial working memory in the treatment group at week 9 following administration of hyperbaric oxygen therapy (HBOT) at 2.0 ATA, five times per week for four consecutive weeks, compared with week 5 after P. gingivalis induction. This finding suggests that HBOT contributes to the restoration of spatial working memory previously impaired by P. gingivalis exposure. These results are consistent with those of Liu et al. (2007), who reported that HBOT administered after hypoxic–ischemic induction in neonatal rats, as assessed by the Morris Water Maze test, enhanced learning ability and spatial memory. urthermore, histological examination indicated a protective effect against brain tissue damage, particularly in the hippocampus [42].

These findings are supported by the theory that hyperbaric oxygen therapy (HBOT) has significant neuroprotective and cognitive effects. HBOT has been shown to reduce the expression of inflammatory molecules, such as TNF-α and IL-1β, which play a role in the disruption of the blood–brain barrier (BBB), as well as to increase the expression of tight junction proteins, such as occludin and claudin-5, which are important for maintaining BBB integrity [43]. Furthermore, in transgenic mouse models, HBOT administration has been reported to reduce amyloid-β accumulation in the hippocampal area, accompanied by significant improvements in cognitive function and overall cerebral blood circulation [28].

HBOT also induces a temporary increase in tissue oxygen levels, which can initially trigger the formation of reactive oxygen species (ROS). However, as an adaptive response, HBOT stimulates an increase in the activity of endogenous antioxidant enzymes, such as superoxide dismutase (SOD), catalase, and glutathione peroxidase, thereby restoring cellular redox balance. This reduction in ROS levels contributes to decreased oxidative damage to neurons, inhibition of NLRP3 inflammasome-based inflammatory activation, and improvement in neural cell integrity, particularly in the hippocampus, which plays a crucial role in regulating spatial memory. Thus, HBOT helps maintain cognitive function by protecting neurons from chronic oxidative stress [44].

In the pretest group compared to the ninth-week group after HBOT administration, differences in the spatial working memory of the experimental animals were still observed. This finding indicates that HBOT can improve the decline in spatial working memory in rats induced with P. gingivalis, but has not yet restored this function to normal levels. These results are inconsistent with the study by Liu et al. (2013), which used an animal model of traumatic brain injury (TBI) and conducted the Morris Water Maze test. That study reported significant improvements in learning ability and memory after 1–2 weeks of HBOT, with spatial memory performance restored to near pre-injury levels [45].

The differences in these results are likely due to several factors that influence the effectiveness of HBOT in restoring spatial working memory in P. gingivalis-induced rats. These factors include age, immune status, and variations in the animal models used in different studies. In Liu et al.'s (2013) study, the model used was an animal model with traumatic brain injury (TBI), whereas this study used an animal model infected with P. gingivalis. The findings of Ilievski et al. (2018) indicate that P. gingivalis infection can cause chronic neuroinflammation, disruption of blood-brain barrier (BBB) integrity, and accumulation of beta-amyloid and tau hyperphosphorylation similar to Alzheimer's disease conditions. Moreover, this infection is associated with irreversible structural damage, such as synaptic loss and neuronal apoptosis [46]. Therefore, although HBOT can improve spatial working memory in P. gingivalis-induced rats, this therapy cannot repair such irreversible damage.

A study conducted by Ding et al. (2018) showed that middle-aged mice had increased expression of proinflammatory cytokines in the brain, while no similar changes were found in young mice. These findings suggest that the aging process may amplify the impact of infection on brain function [47]. Thus, the results support the notion that age and immune status of experimental animals also influence differences in findings across studies.

CONCLUSION

This study shows that Porphyromonas gingivalis infection significantly reduces spatial working memory function, which is thought to be related to the ability of the bacteria to cross the blood-brain barrier (BBB) through its virulence factors, including lipopolysaccharide (LPS) and gingipain, trigger immune dysfunction, and induce neuroinflammation and structural brain damage. For a duration of four weeks, HBOT was provided at a pressure of 2.0 ATA, with sessions conducted five times each week, provided improvements in spatial working memory function, although it did not fully restore it to normal conditions. This therapeutic effect is in line with previous findings that HBOT can reduce the expression of inflammatory molecules and improve BBB integrity, but its effectiveness is influenced by the severity of brain damage and the systemic condition of the test animals. Further research is needed to evaluate molecular parameters more comprehensively and optimize the duration and dose of HBOT in periodontitis-related P. gingivalis-induced neuroinflammatory disorders.

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